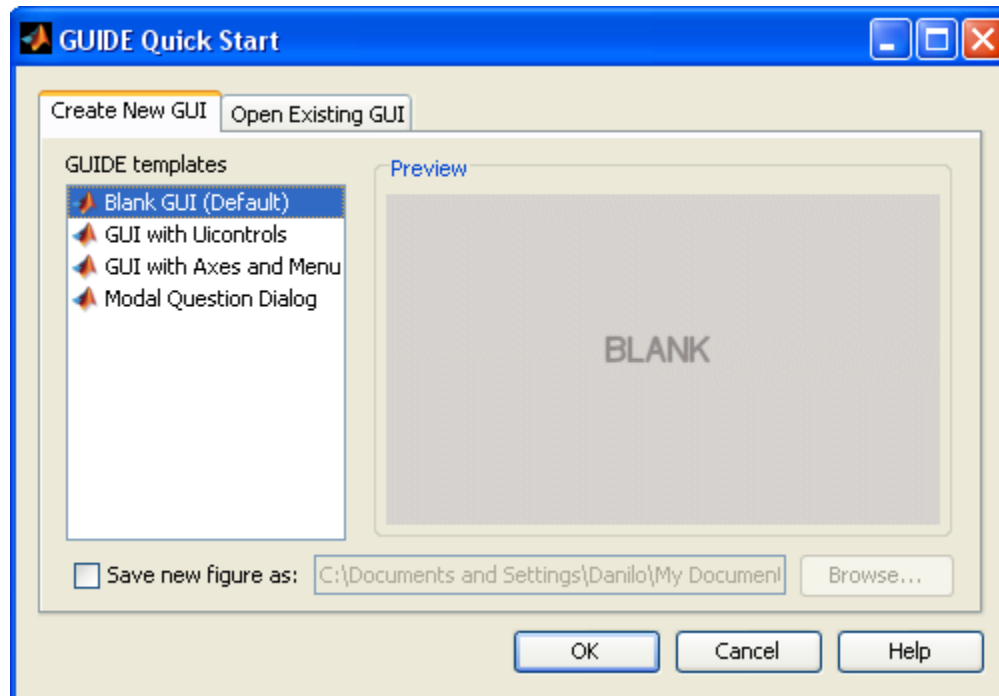


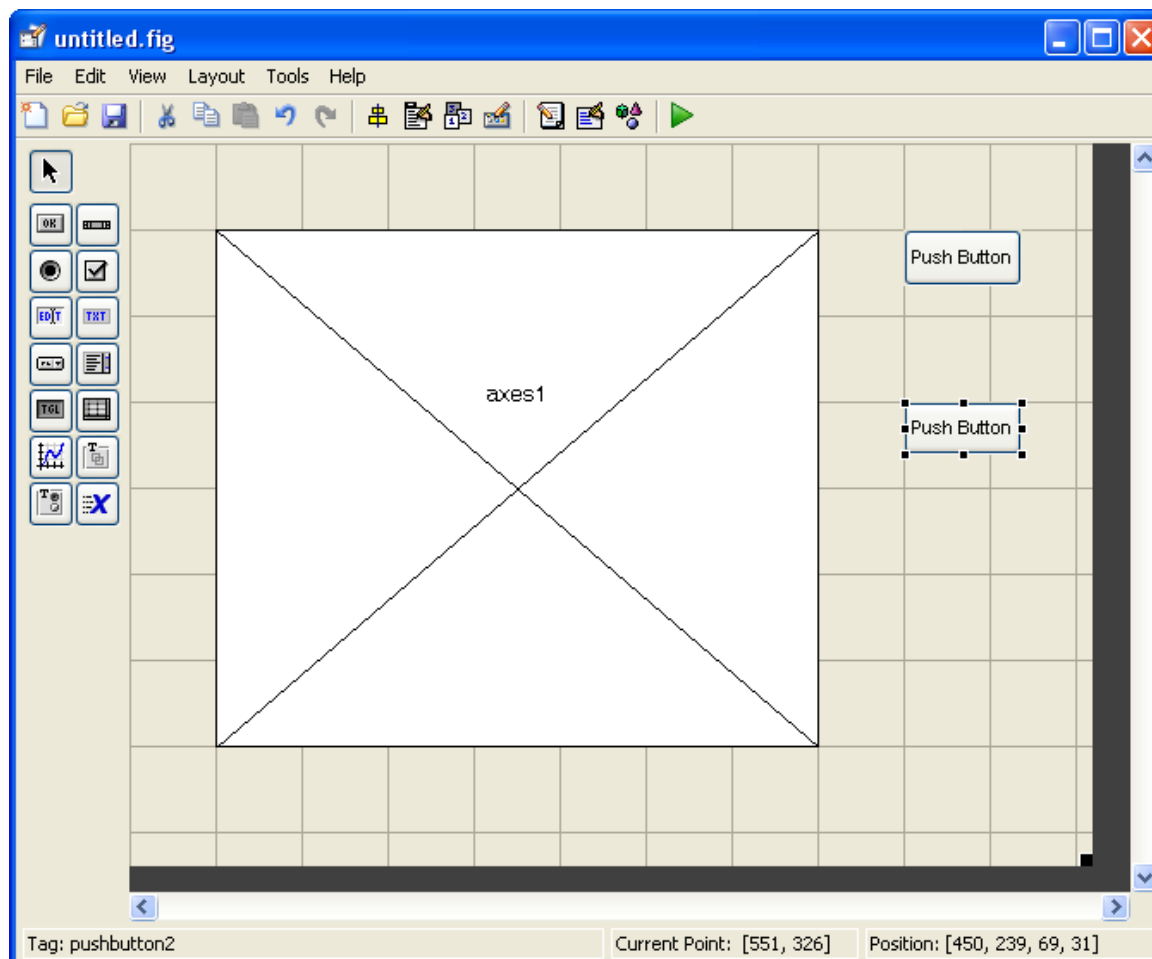
Making GUIs

- It's really easy to make a graphical user interface in Matlab
- To open the graphical user interface development environment, type **guide**
 - » **guide**
 - Select **Blank GUI**



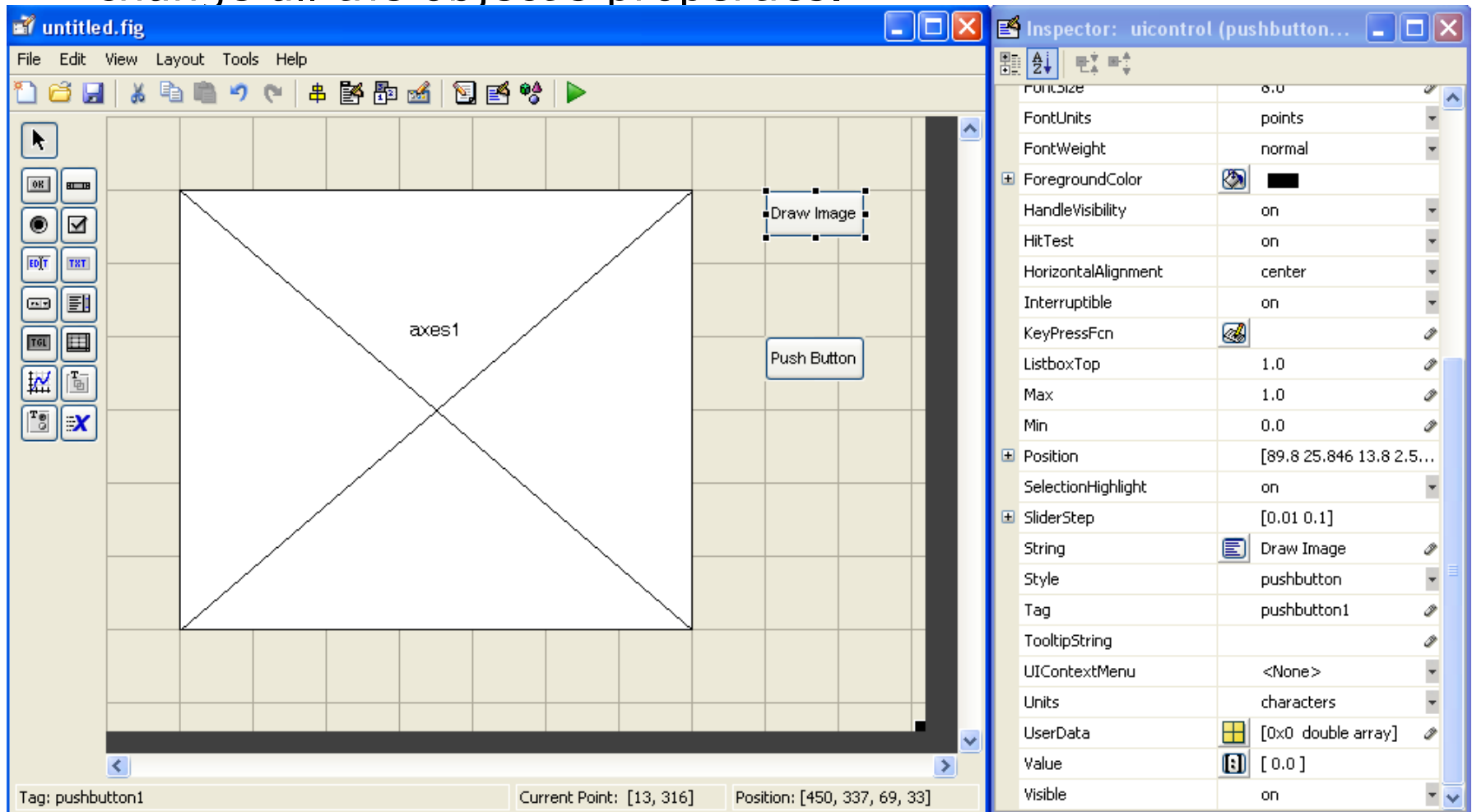
Draw the GUI

- Select objects from the left, and draw them where you want them



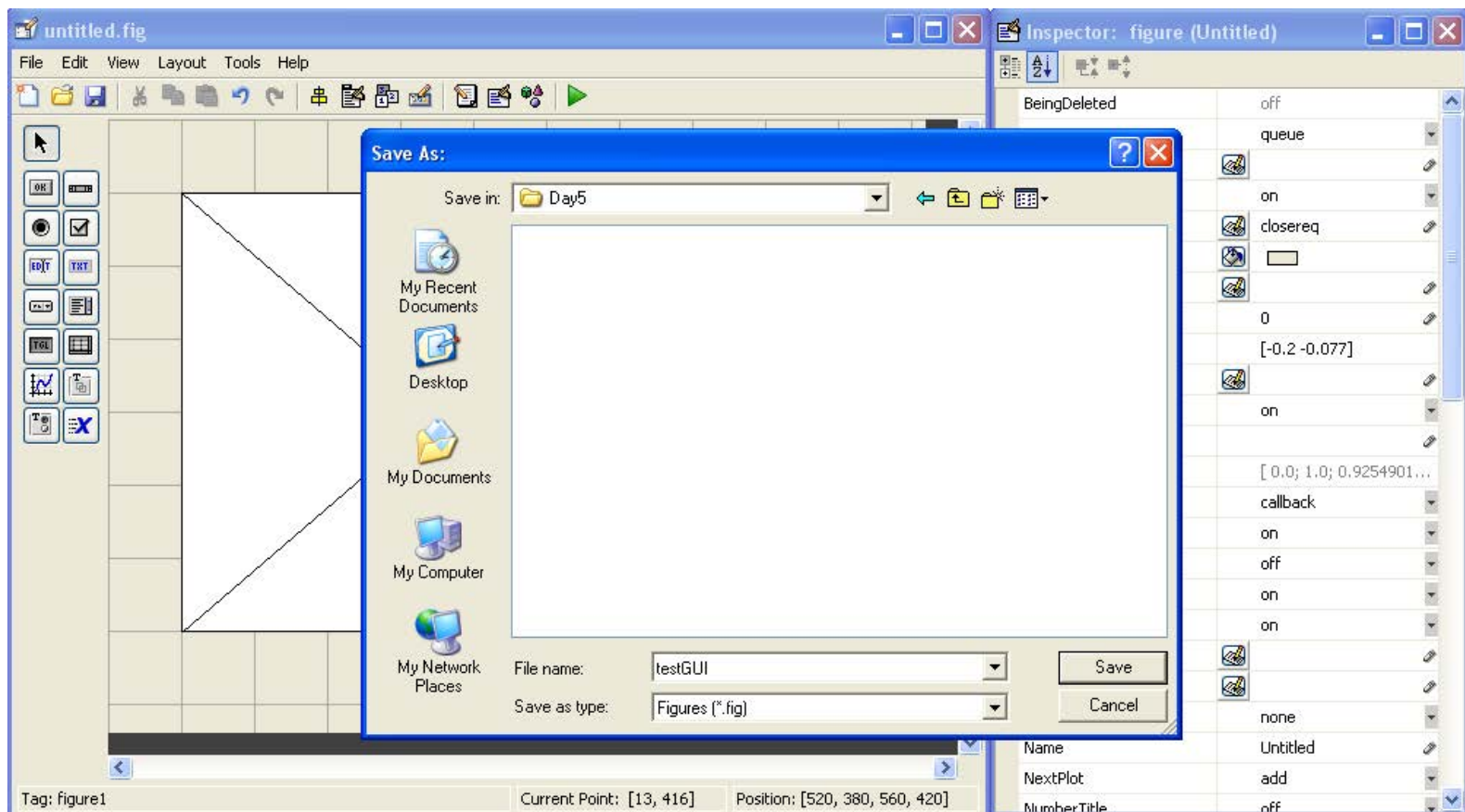
Change Object Settings

- Double-click on objects to open the **Inspector**. Here you can change all the object's properties.



Save the GUI

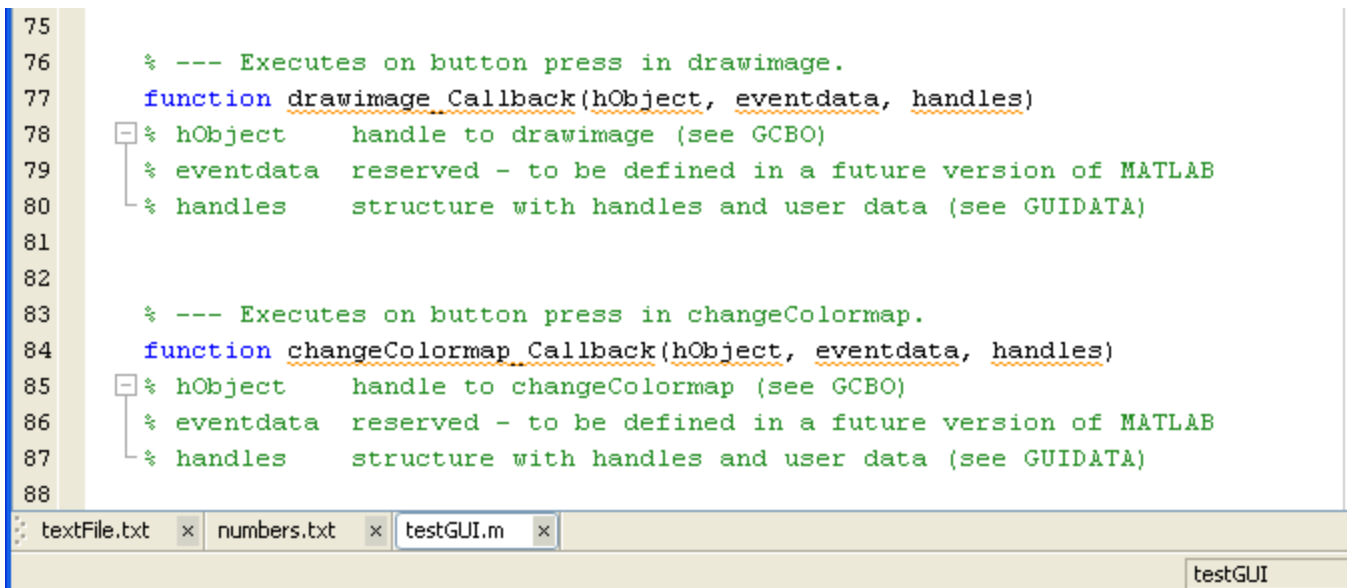
- When you have modified all the properties, you can save the GUI
- Matlab saves the GUI as a .fig file, and generates an m-file!



Add Functionality to M-File

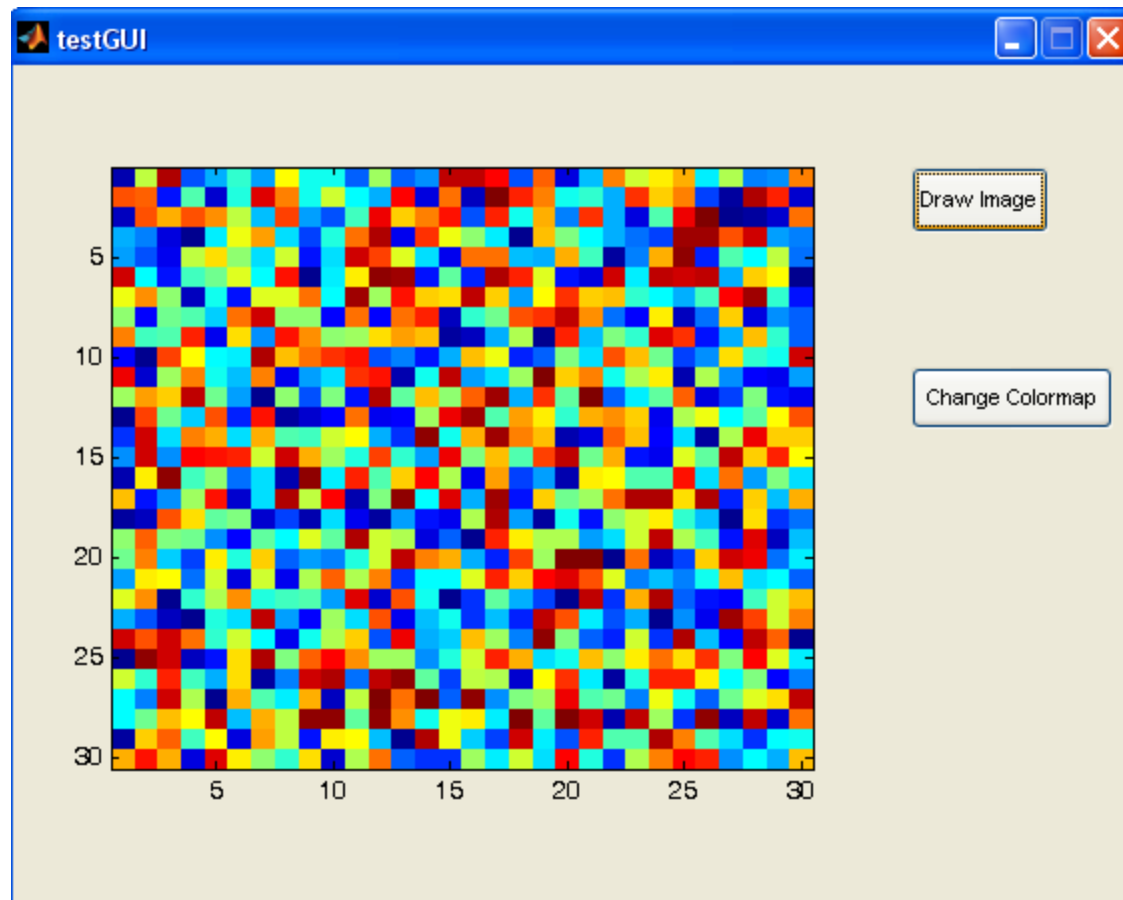
- To add functionality to your buttons, add commands to the 'Callback' functions in the m-file. For example, when the user clicks the Draw Image button, the `drawimage_Callback` function will be called and executed
- All the data for the GUI is stored in the handles, so use `set` and `get` to get data and change it if necessary
- Any time you change the handles, save it using `guidata`
» `guidata(handles.Figure1,handles) ;`

```
75
76     % --- Executes on button press in drawimage.
77     function drawimage_Callback(hObject, eventdata, handles)
78     % hObject      handle to drawimage (see GCBO)
79     % eventdata    reserved - to be defined in a future version of MATLAB
80     % handles      structure with handles and user data (see GUIDATA)
81
82
83     % --- Executes on button press in changeColormap.
84     function changeColormap_Callback(hObject, eventdata, handles)
85     % hObject      handle to changeColormap (see GCBO)
86     % eventdata    reserved - to be defined in a future version of MATLAB
87     % handles      structure with handles and user data (see GUIDATA)
88
```

A screenshot of the MATLAB IDE. The main window displays the 'testGUI.m' file, which contains two callback functions: 'drawimage_Callback' and 'changeColormap_Callback'. Both functions have comments explaining their parameters: 'hObject' is the handle to the button, 'eventdata' is reserved for future MATLAB versions, and 'handles' is a structure containing handles and user data. The file editor shows line numbers from 75 to 88. At the bottom of the window, there is a taskbar with three open files: 'testFile.txt', 'numbers.txt', and 'testGUI.m'. The 'testGUI' window is also visible in the bottom right corner.

Running the GUI

- To run the GUI, just type its name in the command window and the GUI will pop up. The debugger is really helpful for writing GUIs because it lets you see inside the GUI



GUI Helper Functions

- Use keyboard to allow debugging from command window. GUI variables will appear in the workspace. Use return to exit debug mode
 - Use built-in GUI modals for user input:
 - » `uigetfile`
 - » `uiputfile`
 - » `inputdlg`
- And more... (see help for details)